Place value in decimals

Task A: Exponential and fractional forms for decimals

1) Fill in the gaps so that each column shows a different way of writing the same value.

In fractional form	$\frac{1}{10}$	$\frac{1}{100}$	$\frac{1}{1000}$	$\frac{1}{10\,000}$
In decimal form	0.1	0.01	0.001	0.0001
In exponent form	$\frac{1}{10}$	$\frac{1}{10^2}$	$\frac{1}{10^3}$	$\frac{1}{10^4}$
In words	one tenth	one hundredth	one thousandth	One ten thousandth

2) Write the following in decimal form:

a) $2 \times 10^3 + 3 \times 10^2 + 4 + 9 \times \frac{1}{10} =$ 2304.9

2000 + 300 + 4 + 0.9

b) $8 \times 10^3 + 4 + 0.7 + 0.02 =$ 8004.72

8000 + 4 + 0.7 + 0.02

c) $9 \times 10^2 + 6 + 0.8 + 7 \times \frac{1}{10^3} =$ 906.807

900 + 6 + 0.8 + 0.007

d) $2 \times 10^3 + 40 + 0.003 + 4 \times \frac{1}{10} + 6 \times \frac{1}{10^3} =$ 2040.409

2000 + 40 + 0.003 + 0.4 + 0.006

Task B: Different representations of decimals

1) Complete the charts to make the decimals.

100	500	700
10	50	70
1	5	7
0.1	0.5	0.7
0.01	0.05	0.07

751.15

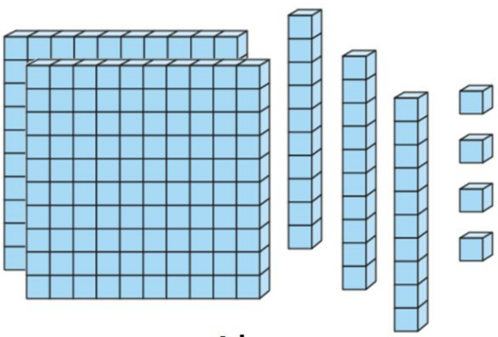
200	600
20	60
2	6
0.2	0.6
0.02	0.06

262.26

10	30	90
1	3	9
0.1	0.3	0.9
0.01	0.03	0.09
0.001	0.003	0.009

93.191

2) Three students have different numbers.



Alex

200	500
20	50
2	5
0.2	0.5

Laura

10^2	10	1	$\frac{1}{10}$	$\frac{1}{10^2}$
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Sofia

Insert all the digits 6, 5, 4 and 2 into Sofia's place value chart to make:

a) a decimal greater than Laura's.

Example answers- 654.2 256.4 645.2

b) a decimal less than Alex's.

Example answers- 54.26 26.45 64.25

c) a decimal in between Alex's number and Laura's number.

Only two answers- 254.6 and 245.6

Try to find more than one answer for each.